BRHANE GIDAY

Portfolio: https://brhane.vercel.app
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GitHub: https://github.com/brhanegiday

Summary

Software & AI Engineer with over 6 years of experience building scalable full-stack applications, robust backend microservices, RESTful APIs, and secure workflow automation platforms across Africa. Proven expertise in Node.js, TypeScript, Python, PostgreSQL, Docker, and Kubernetes, with a strong focus on Digital Public Infrastructure (DPI) and Digital Public Goods (DPGs). Demonstrated track record in architecting high-availability end-to-end systems, integrating open-source frameworks (MOSIP), strengthening platform security, and translating complex public sector requirements into resilient, production-ready software solutions.

Experience

Software & AI Engineer Jan 2025 to Present
Upanzi Network - Kigali, Rwanda

- Architected and deployed high-performance Node.js and Python backend microservices and RESTful APIs, enabling seamless interoperability between legacy Digital Public Infrastructure (DPI) and open-source MOSIP systems.
- Strengthened platform security with OpenID Connect authentication, service isolation and secure credential handling across distributed services.
- Built CI/CD pipelines and containerised services with Docker and Kubernetes for high availability across pan-African research networks
- Work across frontend, backend, DevOps and AI research on DPI projects, including the AI Policy Analyzer and the Fayda National ID chatbot.

Software Engineer Sep 2022 to Nov 2023
Better Mobile Security - Remote, USA

- Engineered secure backend APIs and data integration pipelines for mobile security applications, optimizing data flow and system reliability at scale.
- Collaborated across distributed international teams to enforce strict code quality, robust unit/integration testing standards, and backend security protocols.

Senior Frontend Engineer Mar 2022 to Sep 2022
DoubleDice - Remote, Belgium

- Architected scalable fullstack systems using Next.js, Node.js, and TypeScript with clean backend service abstractions and API integration patterns.
- Integrated Web3 wallet connectivity (MetaMask, WalletConnect) and smart contract interactions to enable seamless crypto transactions
- Led frontend QA, implementing unit tests and end-to-end testing with Jest

Frontend Developer & UI/UX Designer Sep 2020 to Oct 2021
MAD Technologies - Ethiopia

- Delivered client websites efficiently using WordPress and React projects.
- Designed intuitive user interfaces in Figma & Adobe XD.

Skills

- **Backend & Core Dev:** Node.js, RESTful APIs, Microservices Architecture, PostgreSQL, Python, System Design
- **Security & Identity:** DPI, OpenID Connect, OAuth2, Credentials & Key Management, Platform Security
- **DevOps & Infrastructure:** Docker, Kubernetes, Terraform, GitHub Actions, AWS (Certified CCP), Linux (CompTIA+)
- **DPG & Ecosystems:** Digital Public Infrastructure (DPI), Digital Public Goods (DPGs), MOSIP, Workflow Automation
- **AI & Data Engineering:** RAG Pipelines, Vector Databases, OpenAI APIs, FastAPI, Data Pipelines, LLM Orchestration

Education

MSc, Information Technology (Software Engineering & AI) Aug 2023 - May 2025
Carnegie Mellon University Africa, Rwanda

I focused on software engineering, DevOps, and AI with an emphasis on software design, architecture, and scalable systems. My coursework and projects explored distributed systems, cloud-native development, and applied machine learning, combining both research and hands-on practice.

BSc, Electrical & Computer Engineering Sep 2014- Jul 2019
Addis Ababa University, Ethiopia

I specialize in Computer Engineering, with a strong foundation in computer programming, computer architecture, digital logic design, and compilers.

Projects

MOSIP-Integrated Event Management System Backend & Integration Engineer

- Integrated MOSIP open-source identity verification APIs, enabling secure backend biometric and demographic token verification for event authentication.
- Designed secure role-based access control (RBAC) microservices and API endpoints for automated booking and event administration.
- Built a responsive dashboard using Angular Material and NgRx for state management.

OLGA- AI-Powered Academic Advisor Founder & Fullstack Engineer

- Architected and built full-stack AI-powered platform to help students plan academic paths aligned with career goals, integrating responsive client interfaces with secure backend services.
- Implemented CI/CD pipelines, wrote tests, and conducted code reviews.
- Collaborated with advisors and faculty, implementing workflows that streamline student–advisor interactions.

Fayda Chatbot System Backend & AI Systems Engineer

- Architected multi-channel webhook handlers and messaging worker services supporting Telegram and WhatsApp for automated National ID inquiries.
- Implemented support for multiple writing styles, including native Ge'ez script and phonetic Latin-script queries.
- Engineered a grounded RAG data pipeline and asynchronous human-in-the-loop review workflow service for National ID Agency staff verification.

Policy Analyzer Software & AI Engineer

- Engineered an automated policy analysis backend service using RAG pipelines to process unstructured regulatory texts for Digital ID and Data Protection frameworks across African nations.
- Designed interactive visual dashboards for international development stakeholders, enabling data-driven evaluations of Digital Public Infrastructure (DPI) readiness across target regions.

Nabayt- Storytelling Platform Fullstack Engineer

- Built an end-to-end storytelling platform with Next.js, NestJS and PostgreSQL.
- Integrated AI content moderation that scans every post in under 10 seconds before publishing.

KaziData – Labor Market Intelligence platform Fullstack & Data Engineer

- Architected full-stack labor-market data pipelines and backend microservices alongside interactive frontend interfaces to process job postings, curricula, and official labor data.
- Engineered a connected skills graph mapping engine and frontend analytics dashboards for workforce intelligence, skill gap analysis, and program alignment.
- Implemented multi-language localization supporting English, French, and Kinyarwanda across data integration workflows and user interfaces.
- Leading a team of engineers and interns

Certifications

- CompTIA Linux+, December 2024
- AWS Certified Cloud Practitioner, January 2025